# HIV Infection: Detailed Knowledge Document

Paragraph-based reference material for medical information retrieval and RAG chatbot testing

**Educational and safety note:** This document provides general health information and is not a substitute for a qualified healthcare professional. Testing, diagnosis, and treatment decisions should be made with an appropriate health service. HIV is treatable, and people living with HIV can live long, healthy lives with effective care.

## 1. What Is HIV?

Human immunodeficiency virus (HIV) is a virus that attacks the immune system, especially CD4 T lymphocytes, which help the body coordinate its response to infections. If HIV is not treated, it can gradually reduce the number and function of these cells and make it harder for the body to fight certain infections and cancers. HIV is a chronic, manageable condition with modern antiretroviral therapy. HIV and AIDS are not the same: HIV is the virus and infection, while acquired immunodeficiency syndrome (AIDS) is the most advanced stage of HIV infection, diagnosed using specific clinical criteria. Not everyone with HIV develops AIDS, particularly when treatment is started and continued.

## 2. How HIV Is Transmitted

HIV can be transmitted when certain body fluids from a person with HIV enter another person's bloodstream or contact susceptible mucous membranes or damaged tissue. Relevant fluids include blood, semen, pre-seminal fluid, vaginal fluids, rectal fluids, and breast milk. Common routes include anal or vaginal sex without effective prevention, sharing needles or other injecting equipment, and transmission during pregnancy, childbirth, or breastfeeding. The risk can be greatly reduced through effective treatment, prevention medicines, condoms, sterile injecting equipment, and appropriate care during pregnancy and breastfeeding. The details of prevention in pregnancy should be discussed with a qualified clinician.

## 3. How HIV Is Not Transmitted

HIV is not spread through ordinary social contact. It is not transmitted by hugging, shaking hands, sharing food or utensils, using the same toilet, coughing, sneezing, sweat, tears, or mosquito bites. Sharing a home, classroom, or workplace with a person living with HIV does not transmit HIV. Casual contact and intact skin do not provide a route for HIV transmission. Accurate information helps prevent stigma and discrimination against people living with HIV.

## 4. Symptoms and Stages of HIV

Some people develop a short flu-like illness approximately two to four weeks after infection, although timing and symptoms vary and some people notice no symptoms. Possible early symptoms include fever, sore throat, rash, swollen lymph nodes, headache, muscle aches, fatigue, and night sweats. These symptoms are not specific to HIV and cannot confirm infection. After the early stage, a person may have no noticeable symptoms for years while the virus remains in the body. Without treatment, immune damage can progress. Advanced HIV can be associated with severe weight loss, prolonged fever or night sweats, persistent diarrhoea, oral or other fungal infections, tuberculosis, and other opportunistic infections. Symptoms alone cannot diagnose or rule out HIV; testing is necessary.

## 5. HIV and AIDS: The Difference

HIV refers to infection with the human immunodeficiency virus. AIDS refers to the most advanced stage of HIV infection, when severe immune damage or certain defining illnesses are present. In many clinical guidelines, an AIDS diagnosis can involve a CD4 count below 200 cells per cubic millimetre or an AIDS-defining condition. Criteria and clinical interpretation should be assessed by a healthcare professional. Effective antiretroviral therapy can prevent progression to AIDS, improve immune function, and reduce the risk of HIV-related illness.

## 6. HIV Testing and the Window Period

HIV testing is the only reliable way to determine whether a person has HIV. Tests include laboratory antigen/antibody tests, rapid tests, antibody-only tests, and nucleic acid tests that detect viral genetic material. Different tests become able to detect infection at different times, called the window period. A negative result soon after a possible exposure may not exclude infection, so a repeat test may be recommended based on the test type and exposure date. Follow the testing service's advice about timing. A reactive screening result generally needs follow-up testing according to the national testing algorithm before a diagnosis is confirmed. Confidential testing and counselling are available through healthcare services, including public health programmes.

## 7. CD4 Count and Viral Load

The CD4 count measures a type of immune cell and helps clinicians understand the condition of the immune system. A lower count can indicate greater immune damage and may affect decisions about screening or prevention of opportunistic infections. The viral load test measures the amount of HIV RNA in the blood. It is used to assess how well treatment is working. With effective therapy, viral load can become undetectable on standard tests. Clinicians monitor viral load and other health measures over time to guide care. Results should be interpreted in context rather than used as stand-alone self-diagnosis tools.

## 8. Antiretroviral Therapy (ART)

Antiretroviral therapy is the standard treatment for HIV. ART uses a combination of medicines that stop the virus from multiplying. Treatment is recommended for people diagnosed with HIV and should be started as soon as clinically appropriate, in consultation with a qualified provider. Taking ART consistently can reduce viral load, preserve or restore immune function, lower the risk of HIV-related illness, and help people live longer, healthier lives. The specific regimen depends on national guidance, other medical conditions, possible drug interactions, pregnancy considerations, resistance testing where indicated, and individual circumstances. People should not start, stop, or change HIV medicines without professional advice.

## 9. Undetectable = Untransmittable (U=U)

Evidence shows that a person living with HIV who takes treatment and maintains a sustained undetectable viral load does not sexually transmit HIV. This is commonly called Undetectable = Untransmittable, or U=U. Maintaining viral suppression requires continuing treatment and follow-up viral-load monitoring. U=U refers to sexual transmission; it should not be assumed to address every possible transmission route, such as sharing injection equipment or breastfeeding. A healthcare professional can explain how the evidence applies to an individual's circumstances.

## 10. HIV Prevention

HIV prevention can include using condoms correctly, taking pre-exposure prophylaxis (PrEP) when appropriate, and using post-exposure prophylaxis (PEP) after a potential exposure. PEP is time-sensitive and should be started as soon as possible, ideally within 24 hours and no later than 72 hours after exposure, with urgent medical assessment. PrEP is medicine used by HIV-negative people to reduce the chance of acquiring HIV; suitability and testing schedules should be discussed with a healthcare provider. Other prevention steps include never sharing needles or injecting equipment, ensuring sterile equipment for medical or cosmetic procedures, and accessing testing and treatment services. HIV prevention and care during pregnancy can greatly reduce transmission to the baby.

## 11. What to Do After a Possible Exposure

If a person thinks they may have been exposed to HIV, they should seek urgent medical advice rather than wait for symptoms. A clinician can assess the type of exposure, discuss baseline and follow-up testing, and determine whether PEP is appropriate. Because PEP must be started within 72 hours, prompt action matters. Do not take another person's HIV medicines or attempt to create a treatment regimen independently. If the exposure involved sexual assault, seek emergency medical care and support services as soon as possible.

## 12. Pregnancy, Birth, and Breastfeeding

HIV can be transmitted during pregnancy, childbirth, or breastfeeding, but appropriate treatment and clinical care can greatly reduce the risk. HIV testing and early treatment during pregnancy are important. The safest feeding approach can depend on national guidelines, access to safe alternatives, and individual clinical circumstances. Pregnant or breastfeeding people living with HIV should work closely with their healthcare team and should not make treatment or feeding decisions based solely on general online information.

## 13. Living With HIV and Long-Term Health

With effective treatment and ongoing care, many people living with HIV can have a near-normal life expectancy. Long-term care may include regular viral-load monitoring, checking for medication side effects and interactions, vaccination according to clinical guidance, screening for other infections and health conditions, and support for mental health and wellbeing. Healthy eating, physical activity, sleep, and avoiding tobacco can support general health but do not replace ART. Anyone living with HIV should have a care plan tailored to their needs.

## 14. Stigma, Privacy, and Mental Health

HIV is a medical condition, not a measure of a person's character or worth. Stigma and misinformation can discourage people from testing or seeking care. HIV status is private health information and should be handled respectfully and confidentially according to applicable law and clinical policy. Emotional support, counselling, peer support groups, and trusted healthcare professionals can help people cope with diagnosis, treatment, relationships, and disclosure decisions. A person should not be pressured to disclose private medical information to others outside relevant legal or clinical requirements.

## 15. Common Myths and Facts

**Myth:** You can identify HIV by appearance. **Fact:** A person can have HIV without visible symptoms; testing is needed.
**Myth:** HIV spreads through hugs or shared meals. **Fact:** Ordinary social contact does not transmit HIV.
**Myth:** HIV treatment cures the infection completely. **Fact:** ART controls HIV very effectively but does not currently eliminate it from the body.
**Myth:** A person with a sustained undetectable viral load transmits HIV through sex. **Fact:** Sustained viral suppression means there is no sexual transmission of HIV (U=U).

## 16. Frequently Asked Questions

**Can HIV be cured?** There is currently no widely available cure, but ART can control the virus very effectively.
**Can someone with HIV live a long life?** Yes. With early diagnosis, effective treatment, and ongoing care, many people live long, healthy lives.
**Do symptoms confirm HIV?** No. Many conditions cause similar symptoms; a test is needed.
**Can HIV spread by mosquitoes?** No. Mosquitoes do not transmit HIV.
**What is the first step after a possible exposure?** Seek urgent medical assessment, especially if it has been less than 72 hours, to assess whether PEP is appropriate.

## 17. Key Terms for Medical Question Answering

**HIV:** Human immunodeficiency virus. **AIDS:** Advanced stage of HIV infection with severe immune damage or defining conditions. **ART:** Antiretroviral therapy, the standard treatment for HIV. **CD4 count:** A measure of immune-system cells. **Viral load:** Amount of HIV RNA in blood. **Undetectable:** Viral load below the limit detected by a particular test. **PrEP:** Medicine taken before potential exposure to prevent HIV. **PEP:** Emergency medicine taken after a potential exposure to prevent HIV; it must be started within 72 hours. **Window period:** Time after exposure during which a test may not yet detect infection.

## 18. Suggested Questions for RAG Testing

What is HIV and which cells does it affect? How is HIV transmitted? Can HIV spread through hugging or sharing food? What are early symptoms of HIV? Why can symptoms not confirm HIV infection? What is the difference between HIV

and AIDS? What is the HIV testing window period? What is the difference between CD4 count and viral load? What is antiretroviral therapy? What does U=U mean? What are PrEP and PEP? How soon must PEP be started after a possible exposure? Can a person with HIV live a long life? How can HIV transmission during pregnancy be reduced? What are common myths about HIV?

## 19. Safety and Scope

This document is intended as general educational reference text for information retrieval and chatbot testing. It does not replace current local clinical guidelines or personalised medical advice. HIV testing, treatment, prevention medicines, and follow-up should be provided through qualified health services. For a possible exposure within the last 72 hours, seek urgent care to assess PEP. For other urgent health concerns, contact local emergency or medical services.